

## Lab 02 – First Contact with OpenPLC

Industrial Security (Bachelor)

• Maps to Section 02 • 30–45 min

LAB 02

### 🎯 Goal

Bring up the lab Docker stack and confirm that the soft PLC accepts and runs a ladder-logic program. End the lab with a Modbus read working against it.

### ✂ Step 1 – Bring up the stack

```
1 | cd bachelor/labs/_stack
2 | docker compose up -d
3 | docker compose ps           # all services should show Up
```

### ✂ Step 2 – Open the lab landing page & the OpenPLC web UI

Browse to <http://127.0.0.1:8000> for the lab home page (overview of every lab and the stack). Then open <http://127.0.0.1:8080> for the OpenPLC web UI; log in with the default OpenPLC credentials (openplc / openplc) and change the password right away. Record the new password in your notes.

### ✂ Step 3 – Upload the ladder program

Upload the file `bachelor/labs/_stack/scripts/ladder/conveyor.st` (host-relative path; pick it from your filesystem in the upload dialog). Click **Start PLC**. Open the **Monitoring** page and watch **Motor** toggle when you flip **Start**. Note the cycle time shown at the bottom of the page.

### ✂ Step 4 – Read holding registers

```
1 | docker compose exec student python3 /labs/scripts/modbus_read.py
```

You should see ten register values. They will all be zero unless the program has written to them. Record the first three values in your notes.

### 📁 Deliverable

Hand in: (a) a screenshot of the OpenPLC *Monitoring* page with your running program, (b) the terminal output of `modbus_read.py`, and (c) the observed PLC cycle time.

### ⚠ Caution

The Modbus listener is bound to `127.0.0.1:502` on your host. Do not expose it to your network. The OpenPLC default credentials are well known – treat them as already compromised until you change them.